

SPORTIFY HUB

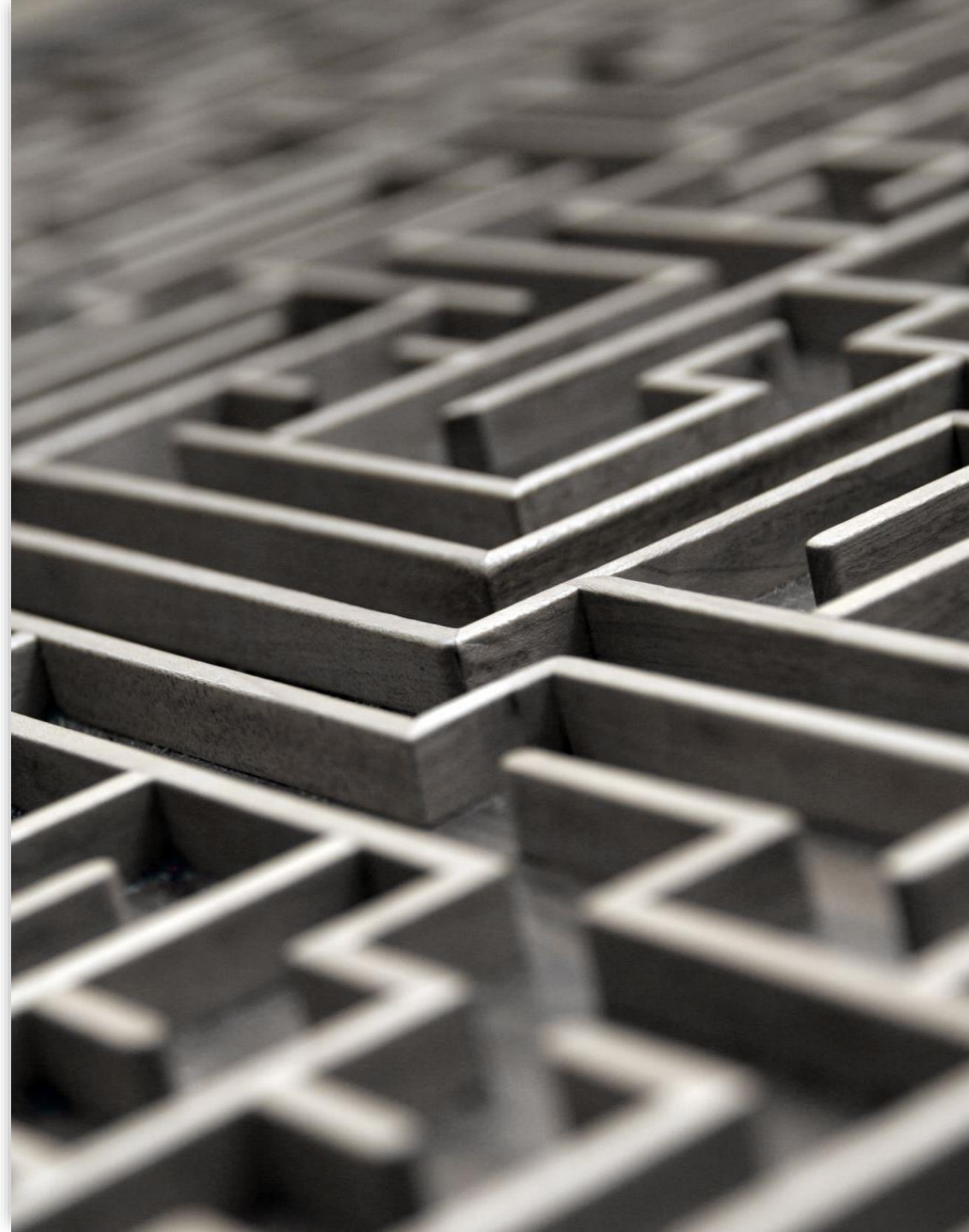
A FUZZY-LOGIC EXERCISE RECOMMENDATION SYSTEM

Final presentation in Fuzzy Sets and Systems II

Fribourg, 31.05.2024

ABSTRACT

Sportify Hub leverages fuzzy logic to provide personalized exercise recommendations tailored to user preferences and goals. The system aims to eliminate the limitations of traditional crisp logic-based recommendation softwares, by mimicking human decision-making processes. While incorporating factors in a relative way, such as age, user goals, available time, and preferred sports, our platform offers a flexible and intuitive way to promote physical activity. This presentation explores the design, implementation, and the origin of our idea in a fuzzy logic-based approach.





INTRODUCTION TO THE PROBLEM

- **Modern lifestyle challenges**

Today's lifestyle and busy schedules make it difficult for individuals to engage in regular physical activity. Many people lack motivation because they struggle to find exercises that fit their specific preferences, or their schedule and place to exercise in general.

- **Existing solutions**

Current exercise recommendation systems often rely on crisp logic, providing generic and rigid suggestions that may not suit everyone. These systems lack adaptability, as well as capturing diverse user requirements.

- **Our approach**

Sportify Hub uses fuzzy logic to create a more personalized(relative) recommendation system, considering few factors that influence an individual's exercise choices.

HISTORICAL CONTEXT AND HUMAN-NATURE CONNECTION

- **Seeking wisdom from the past**

In medieval times, societal roles and the activities of one individual were deeply connected to natural rhythms and personal capabilities.

Example of a natural rhythm: The vikings did not use exact years to date events, a so-called absolute chronology. Instead, they used a relative(fuzzy) chronology with a reference to the number of cozy or cold winters after important events for them. One viking could tell the year by stating “five cold winters after the Battle of Svolder” and that highlights the human connection to nature. Or, if you asked him when he was born, he would tell you "around thirty cold winters ago".

- **Relevance to 'Modern Times'**

By embracing such historical insights, we can design systems that respect individual variability and natural human tendencies. Sportify Hub incorporates philosophies by allowing users to define their age range, free time, fitness goals and favorite sport flexibly.

- **AI as a facilitator**

AI can help us build environments that honor our history and natural inclinations, creating a bridge between past wisdom and present technology.

PROBLEM STATEMENT

- **Crispy age-range and inflexible free-time recommendations**

Most of these systems give an exact choice of age, (a person being aged 46 for example), the recommendations are generic and appointment scheduling is strict.

- **User engagement in activities**

Lack of freedom and choice can lead to reduced user engagement and motivation, as individuals may not find the recommendations relevant or appealing.

- **Health impact**

Inadequate physical activity recommendations or programs can contribute to poor health outcomes, highlighting the need for a more adaptive and personalized approach.



HOW WAS LIFE BEFORE? WE COULD NEVER KNOW

- Hipparchos proposed dividing the day equally into 24 hours which came to be known as equinoctial hours. The concept is based on 12 hours of daylight and 12 hours of darkness on the days of the Equinoxes. However, humans continued to measure time according to seasons for a long period. The system we use today became commonly accepted after the invention of mechanical clocks in the 15th century(at the end of the medieval times).
- Let's be honest. A second is a second, and a minute is a minute ever since. It is a fact that the invention of the mechanical clocks changed the perception of human brains into more structured crispy way rather than the biological one. Christiaan Huygens made the first pendulum clock, regulated by a mechanism with a "natural" period of oscillation in 1656. Galileo studied pendulum motion as early as 1582, but his design for a clock was not built before his death.
- Humans kept track of their physical activities back then without a precise date and time. The difference is, they didn't have the sports that we invented as activities in the past two centuries, neither the mechanical clocks.
- Nowadays it is not possible to show up on a meeting or a conference without checking your clock constantly, or just be on time at your job or lectures in general. We get involved in this system ever since entering kinder-garden or primary school. But is it necessary to stress if you miss your exercise appointment? Isn't the point of exercise to reset your mind and release the stress that you have from today's society?



OUR HUMAN-BASED APPROACH IN THE APPLICATION

- **Age classification**

Users can classify their age in a way that resonates with their self-perception, such as "young", "young turning into adult", "adult ", "adult getting old" and "old", accommodating a wide range of opinions.

- **Free time & favorite sport fuzzy-logic implementation**

We offer the option to schedule appointments in intervals, instead of exactly-timed training sessions, which is limiting. Also, favorite sport can be chosen between multiple different categories: team sports, individual sports, adventure sports, racquet sports, combat sports and so on.

- **Goal-oriented recommendations**

By considering user goals like weight loss, muscle building, stamina increase and so on, Sportify Hub produces recommendations to align with individual aspirations.



JAVA(SPRING BOOT) WEB APP PROTOTYPE

Register

Age:

Height:

Weight:

Gender:

Goal:

Favorite Sport:

Welcome, admin

How much time do you have to exercise today?

Exercise Recommendations

You have less than a half hour to exercise today and your goal is Slightly lose weight.

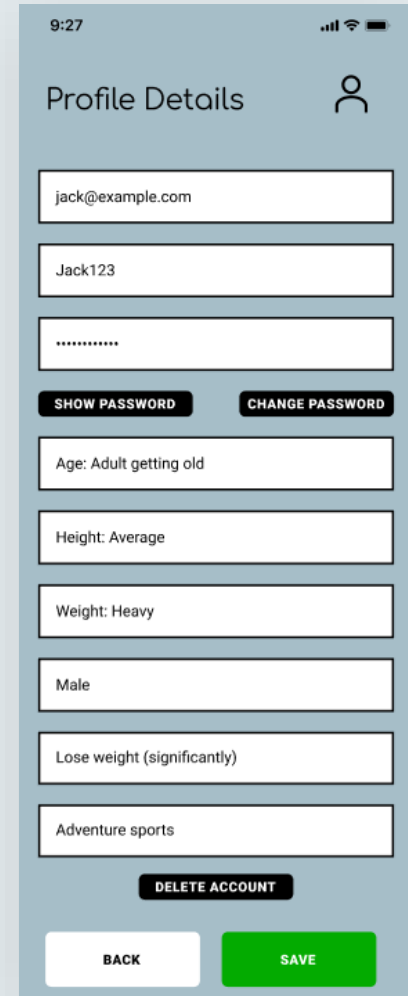
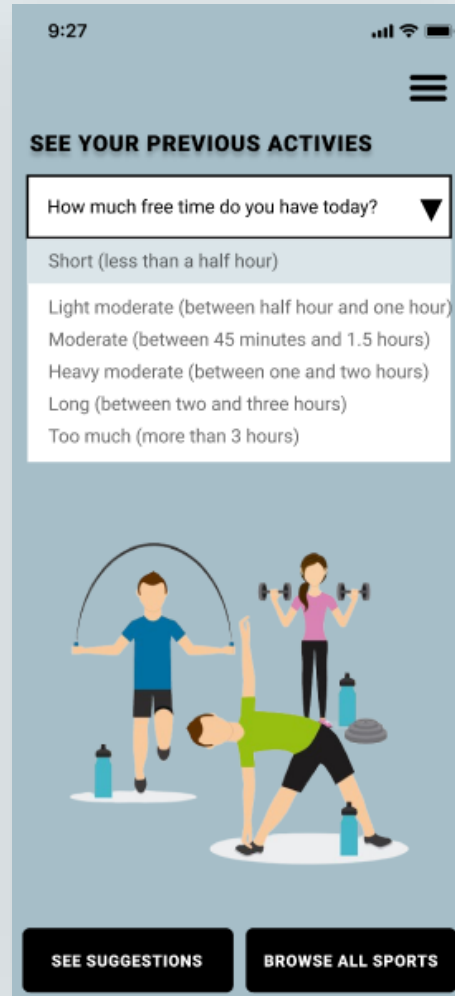
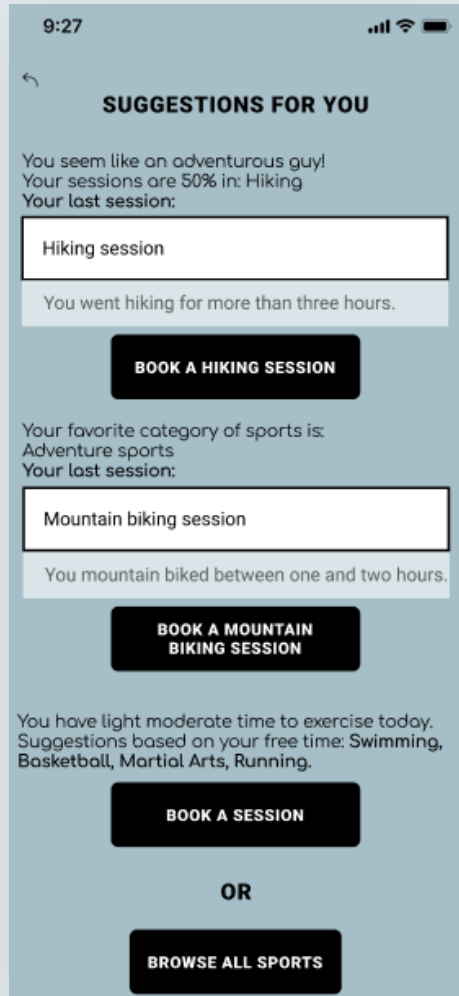
Individual Sports

Running

Cycling

Yoga

MOBILE APP PROTOTYPE(DESIGN)



COMPARISON WITH CRISP LOGIC EXERCISE-BASED SYSTEMS

Criteria	Existing Crisp-Logic Systems	Our Fuzzy-Logic System
Decision Making	Binary decisions based on fixed criteria	Flexible decisions based on a range of inputs
Age Range Selection	Strict age groups (e.g., 18-25, 26-35)	Flexible ranges (e.g., young, young turning adult)
Fitness Goals	Specific goals with strict adherence	Adaptive goals with fuzzy boundaries (e.g., weight loss, muscle building)
Favorite Activities	Predefined, non-intersecting categories	Broad categories with overlap (e.g., individual sports, team sports)
User Personalization	Limited personalization, static profiles	Dynamic profiles adapting to user preferences and changing goals
Exercise Recommendations	Fixed routines based on user profile	Adaptive routines considering user's available time and varying inputs
User Experience	Predictable but limited in scope	Intuitive, adaptable, and user-friendly

EVALUATION



- **Question 1:** Does the user have the option to recommend a sport to similar users?
 - **Answer 1:** As of now, it is not implemented yet. However, it is definitely a good idea, and we are working on improving our application in the future.
-
- **Question 2:** Is the user able to rate the sport after having a session, so he can get better recommendations?
 - **Answer 2:** There is a quick survey after completing the session, just to see if the user enjoyed his exercise experience, and should the system recommend something similar in the future.

CONCLUSION

Sportify Hub's focus is on the potential of using fuzzy logic to create a more human-centric and adaptive exercise recommendation system. By incorporating user preferences, goals, and historical insights, we offer a flexible and personalized approach to promoting physical activity. The main goal is to enhance user engagement and contribute to better health outcomes. As we continue to refine our system, we envision a future where technology integrates with our natural human inclinations, fostering a healthier world.



REFERENCES

- https://www.vikingsofbjornstad.com/Viking_Calendar.shtm#:~:text=The%20Vikings%20did%20not%20use,after%20the%20Battle%20of%20Svolder%E2%80%9D.
- <https://thevikingdragon.com/blogs/news/what-festivals-did-the-vikings-celebrate#:~:text=The%20Vikings%20broke%20the%20year,to%20make%20a%20leap%20year>.
- <https://nrich.maths.org/6070#:~:text=Hipparchus%20proposed%20dividing%20the%20day,hours%20for%20a%20long%20time>.
- Our [mobile app](#) prototype link (just the design in Figma)
- Our [web app](#) GitHub repository (yet to commit and improve)



THANK YOU 🤗

ANY QUESTIONS?